

Erdős Problem #852

EP-852

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $d_n = p_{n+1} - p_n$, where p_n is the n th prime. Let $h(x)$ be maximal such that for some $n < x$ the numbers $d_n, d_{n+1}, \dots, d_{n+h(x)-1}$ are all distinct. Estimate $h(x)$. In particular, is it true that $h(x) > (\log x)^c$ for some constant $c > 0$, and $h(x) = o(\log x)$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>